

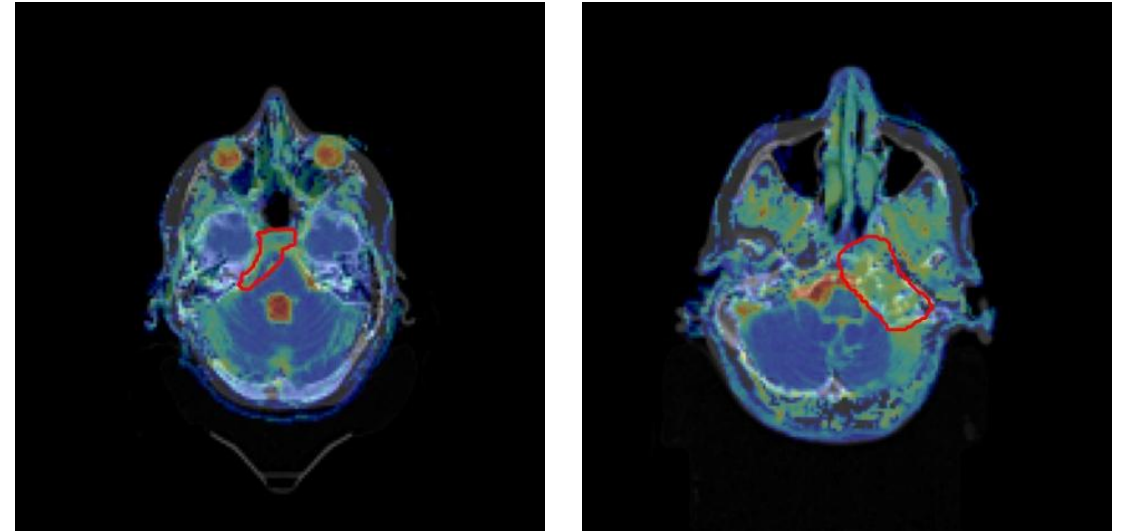
Predicting Tumor Relapse from Anatomical and ADC-Derived Radiomics Biomarkers in Brain Tumor Patients

MSc Semester Project

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Treatment response prediction

- Response to radiation therapy varies considerably
- Accurate treatment response prediction helps clinical decision making
- TCP models often lack true patient specificity
- Survival models often do not include information from medical images

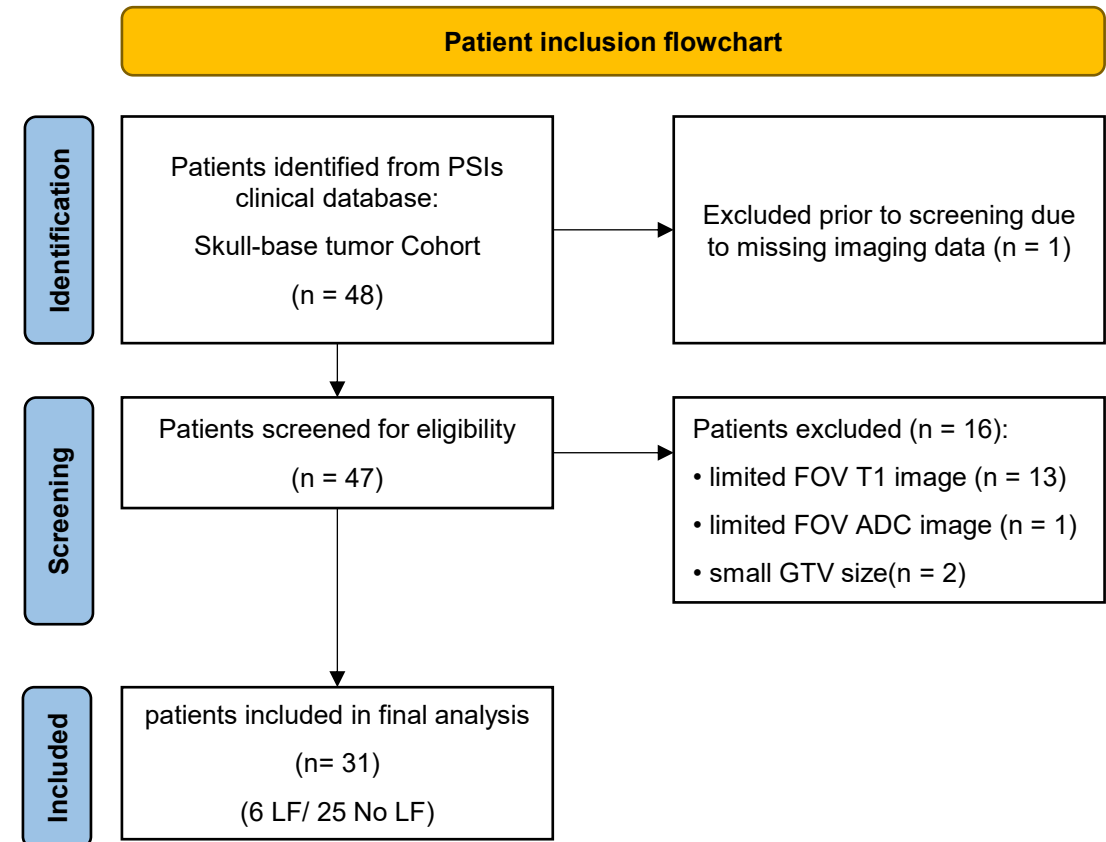


Goal

Investigate the relationship between radiomics features derived from **baseline** imaging and time to local failure after proton therapy

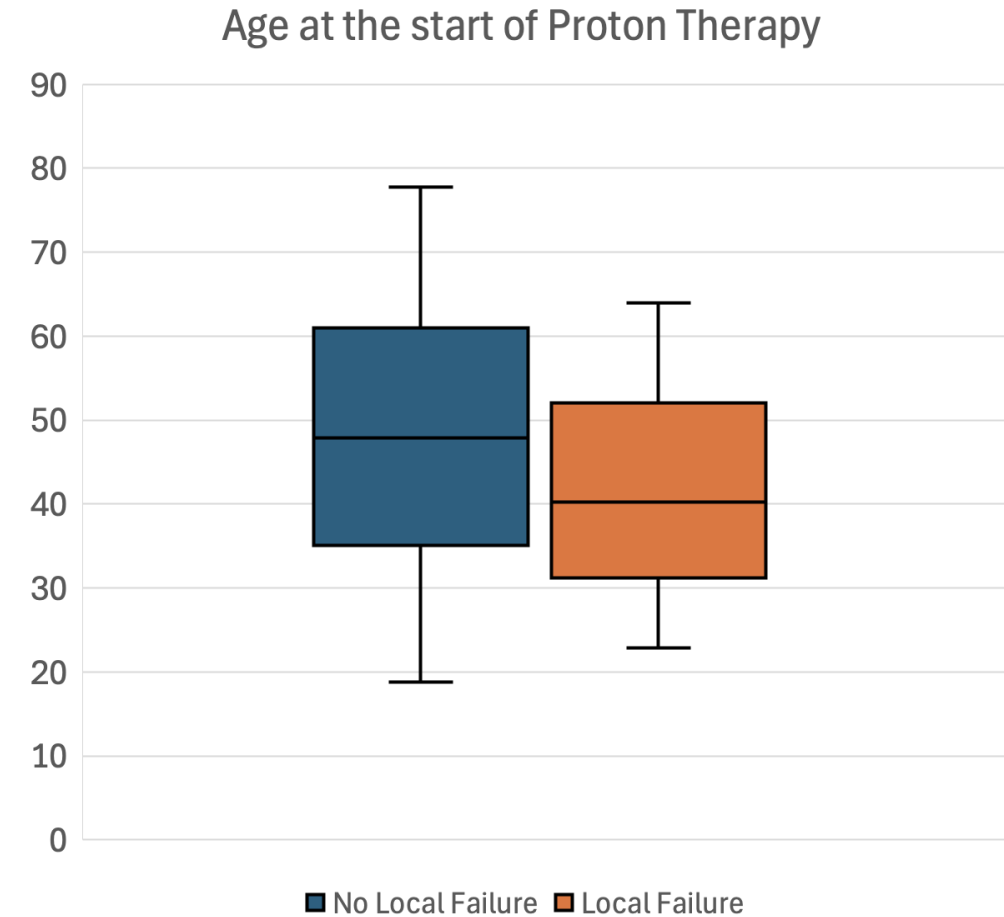
Clinical Data

- Chordoma and Chondrosarcoma treated with PT at PSI between 2015 and 2018
- Baseline ADC(DWI), T1-weighted and FLAIR-weighted MRI images available
- Long-term follow-up data available
- No dose information considered



Clinical Data

- Strong imbalance between sexes
- Median GTV volumes are comparable between local failure and non-failure groups
- Age distributions largely overlap between the groups

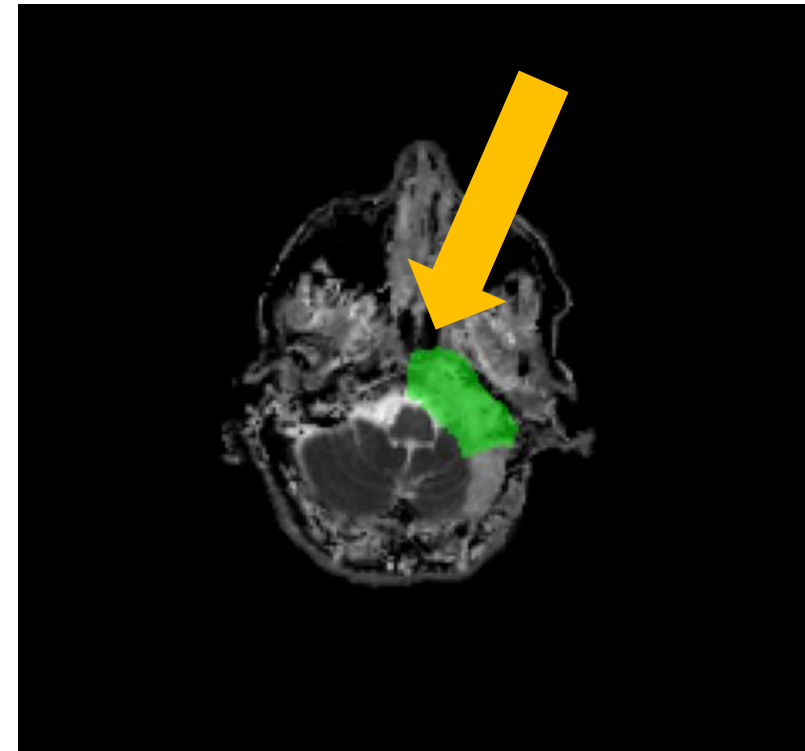


Registration

- Planning CT scan as reference frame
- Enables use of contour data delineated on the CT
→ consistent extraction of radiomics features
- Allows the inclusion of dose information in future projects

GTV Masking

- Binary GTV masks to extract imaging data
- **Exclusion of non-soft-tissue structures**



Normalization

- MRI typically provide relative signal intensities, which limits their power for quantitative analysis
- Normalization using robust z-scoring:

$$ADC_{normalized} = 0.6745 \cdot \frac{ADC - ADC_{median}}{MAD}$$

$$MAD = Median(|ADC - ADC_{median}|)$$

- **FLAIR and T1 normalization using normal appearing white matter masks**



	Median		Skewness		Kurtosis	
	No LF	LF	No LF	LF	No LF	LF
Median ADC	5.164	2.889	0.102	0.685	-0.113	0.448

ADC

- Patients with local failure show lower median ADC values
- Skewness and Kurtosis elevated in LF group

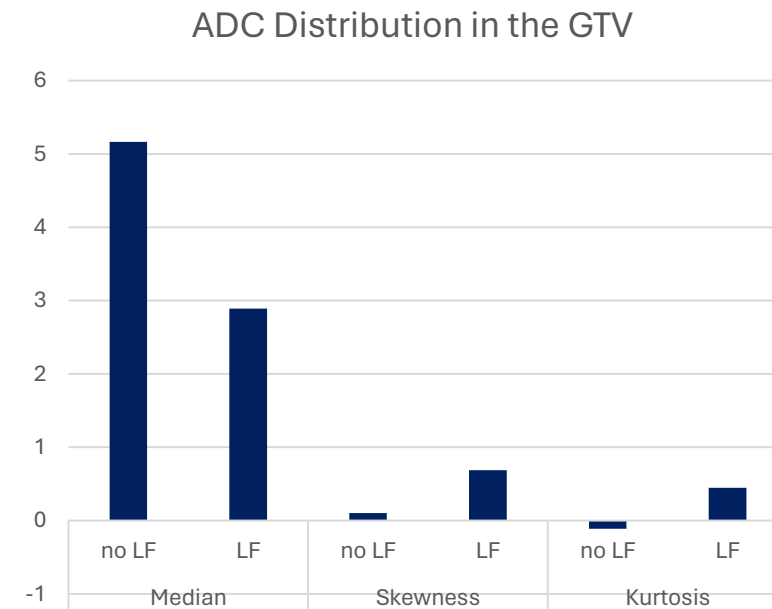


Image Intensity Distributions in the GTV

	Median		Skewness		Kurtosis	
	No LF	LF	No LF	LF	No LF	LF
Median ADC	5.164	2.889	0.102	0.685	-0.113	0.448
Median FLAIR	0.124	0.542	-0.165	-0.267	-0.527	-0.551

ADC

- Patients with local failure show lower median ADC values
- Skewness and Kurtosis elevated in LF group

FLAIR & T1

- Higher median intensities in the LF group (both modalities)
- No consistent differences in distribution shape

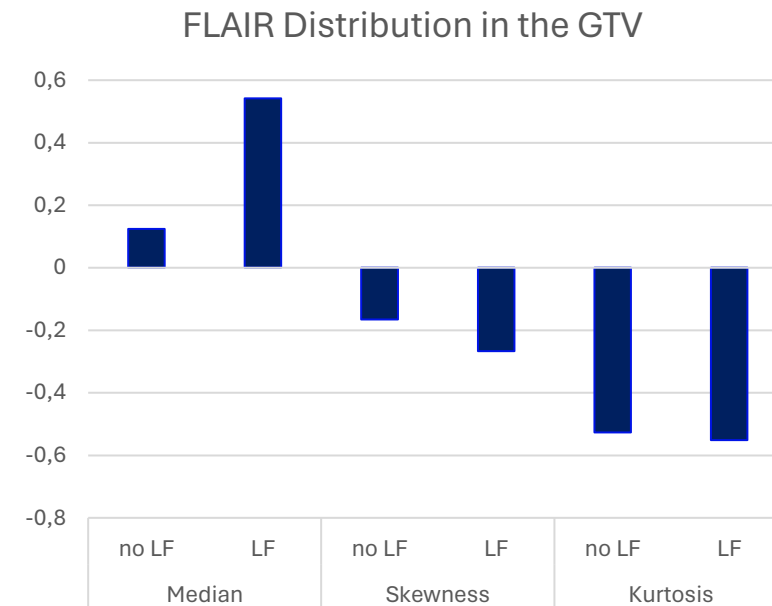


Image Intensity Distributions in the GTV

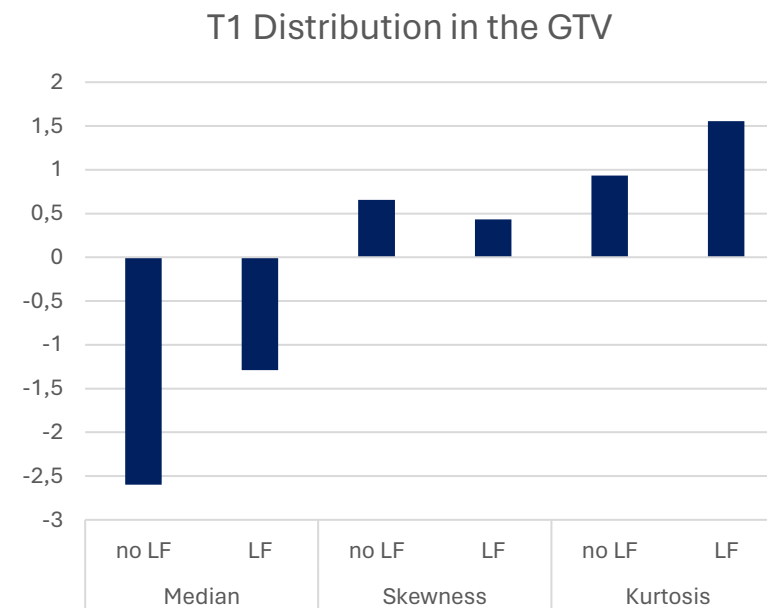
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Median ADC	5.164	2.889	0.102	0.685	-0.113	0.448
Median FLAIR	0.124	0.542	-0.165	-0.267	-0.527	-0.551
Median T1	-2.600	-1.290	0.655	0.434	0.935	1.555

ADC

- Patients with local failure show lower median ADC values
- Skewness and Kurtosis elevated in LF group

FLAIR & T1

- Higher median intensities in the LF group (both modalities)
- No consistent differences in distribution shape

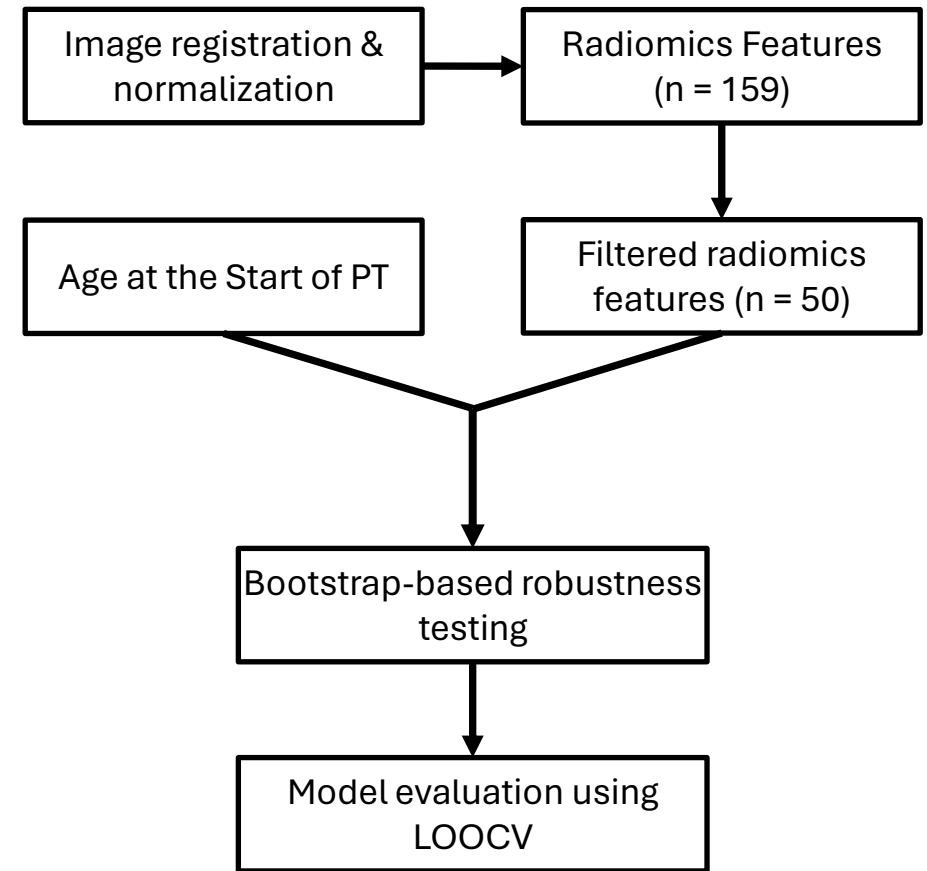


Radiomics Features

- Extraction using PyRadiomics: first order and texture features
- 53 radiomics features extracted per imaging modality (ADC, T1, FLAIR)
- Features filtered for collinearity (Spearman > 0.8)
- Final feature set: 50 radiomics features + age at the start of PT

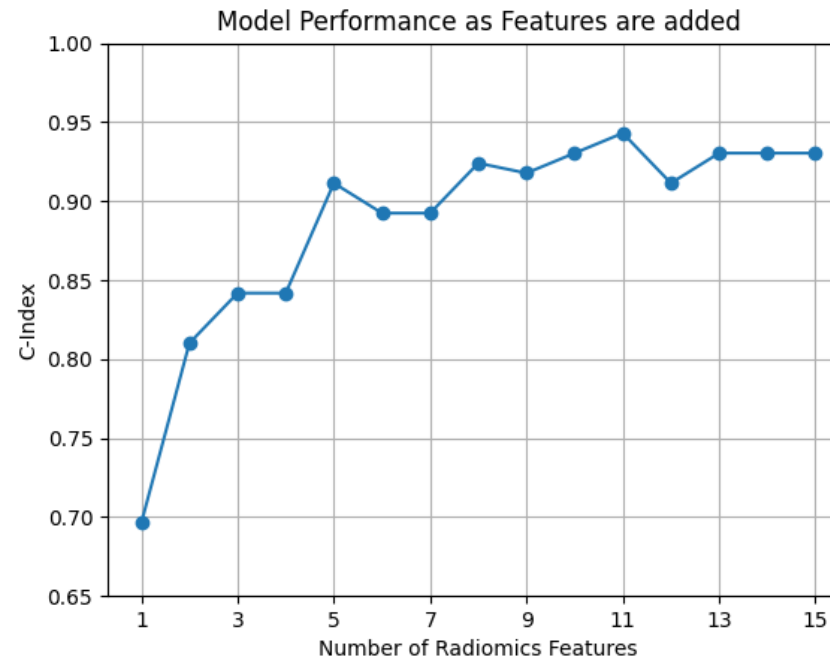
Survival Analysis

- Cox Proportional Hazards model
- Feature robustness assessed using 2000 bootstrap samples



Survival Analysis - Results

- **C-Index:** Increases substantially up to 5 parameters, plateaus after
- **Selection Frequency:** The first 4 parameters show high robustness
- **Integrated Brier Score:** Lowest for 2 parameters, remains comparably low up to 5 parameters



The strongest predictors capture ADC distribution shape and FLAIR signal intensity

#	Feature Added	Selection Frequency	IBS
1	ADC_firstorder_Skewness	0.569	0.078
2	FLAIR_firstorder_90Percentile	0.524	0.077
3	FLAIR_firstorder_Minimum	0.330	0.089
4	ADC_firstorder_Kurtosis	0.313	0.100
5	FLAIR_gldm_DependenceEntropy	0.225	0.089
6	T1_glszm_LowGrayLevelZoneEmphasis	0.208	0.084
7	glszm_LargeAreaEmphasis	0.181	0.109
8	gldm_DependenceNonUniformityNormalized	0.176	0.109
9	Age_at_PT_years	0.145	0.117

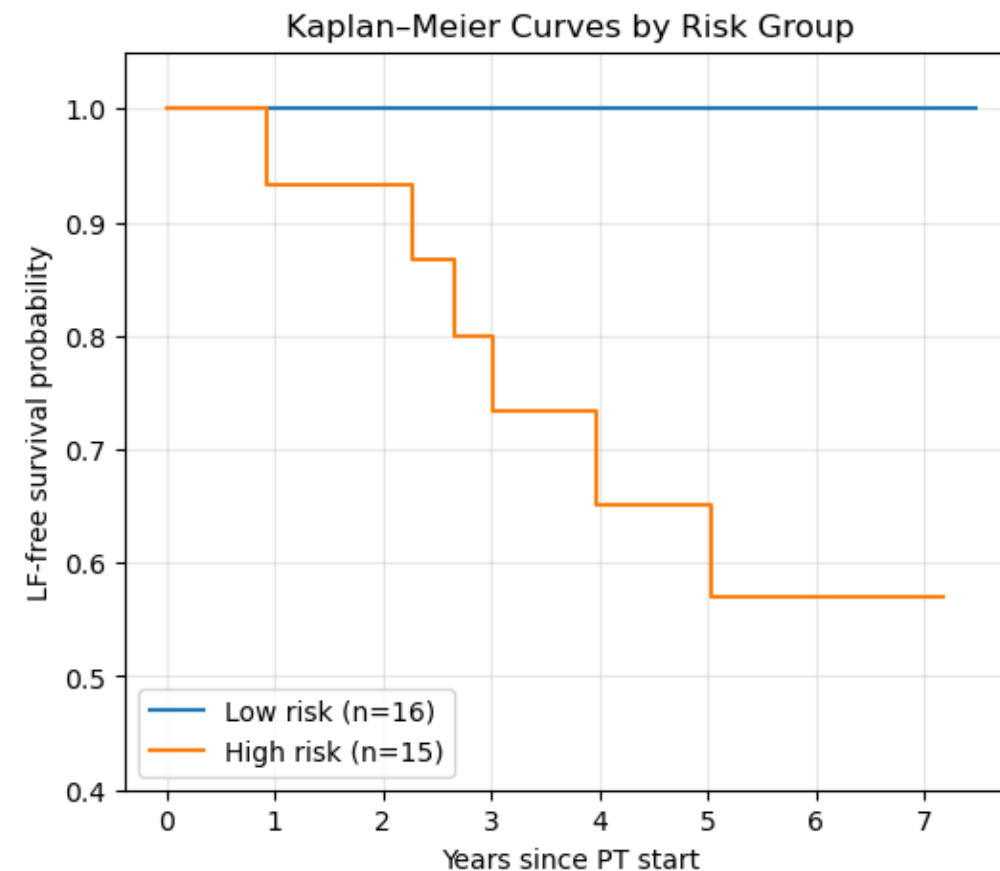
Feature	Coefficient (β)	Hazard Ratio (HR = e^β)
ADC_firstorder_Skewness	0.683	1.979
FLAIR_firstorder_90Percentile	0.762	2.142
FLAIR_firstorder_Minimum	0.495	1.641
ADC_firstorder_Kurtosis	0.442	1.555
FLAIR_gldm_DependenceEntropy	1.232	3.429

Model Parameters

- Positive coefficients for all parameters

Kaplan Meier Analysis

- Curves show good separation between risk groups



→ Selected features capture meaningful biological information

Image Processing

- Feature extraction was improved using air and bone masks
- Reliable intensity normalization using white matter reference tissue

Survival Modeling

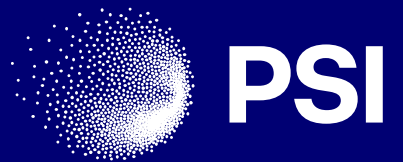
- Model incorporating 5 parameters achieves the best balance between performance and the risk of overfitting
- Kaplan-Meier curves show good risk stratification

Most Predictive Features

- Median ADC values and ADC distribution shape
- FLAIR signal intensity and FLAIR texture

Limitations

- No dose information included
- Small cohort size reduces statistical power
- External validation on an independent test set needed



Thank you for listening!

L. Ackermann
PSI, December 2025